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3.1 Ratio and Proportion

OCR GCSE Mathematics J560 | Component 3 (Ratio, Proportion and Rates of Change) | Papers 1, 2 & 3

3.1.1 Simplifying and Using Ratios

Ratio — a ratio compares two or more quantities of the *same type* and shows their relative sizes. It is written using a colon, e.g. $3:5$.

Writing a Ratio

- Quantities must be in the **same units** before forming a ratio.
- Order matters: "the ratio of boys to girls is $2:3$" is different from $3:2$.
- Ratios have **no units** — they are pure numbers once units are matched.

Worked Example: Write the ratio 40 cm to 1.2 m in its simplest form.

Step 1: Convert to the same unit: $1.2\,\mathrm{m} = 120\,\mathrm{cm}$
Step 2: Write the ratio: $40:120$
Step 3: Simplify by dividing both by HCF (40): $1:3$

Simplifying to Simplest Form

- Divide all parts of the ratio by their **Highest Common Factor (HCF)**.
- A ratio is in simplest form when all parts are integers with HCF = 1.
- For ratios with fractions or decimals, multiply through to clear them first.

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Worked Example: Simplify the ratio $1.5 : 4.5$.

Multiply both by 2 to clear decimals: $3 : 9$
Divide by HCF (3): $\mathbf{1 : 3}$

Equivalent Ratios

Equivalent ratios are found by multiplying or dividing all parts by the same non-zero number, just like equivalent fractions.

Ratio	Multiply by 2	Multiply by 5
$1 : 3$	$2 : 6$	$5 : 15$
$2 : 5$	$4 : 10$	$10 : 25$

Exam Tip: Always check units match before writing a ratio. A common mistake is to write "30 minutes to 2 hours" as $30 : 2$ instead of $30 : 120 = 1 : 4$.

3.1.2 Dividing in a Ratio

Dividing in a ratio — split a total quantity into parts proportional to a given ratio.

Method

1. Find the **total number of parts** (add the ratio values).
2. Find the **value of one part** by dividing the total by the number of parts.
3. Multiply by each ratio value to find each share.

Worked Example: Share £420 in the ratio $3 : 4$.

Step 1: Total parts = $3 + 4 = 7$
Step 2: Value of 1 part = $£420 \div 7 = £60$
Step 3: First share = $3 \times £60 = £180$; Second share = $4 \times £60 = £240$
Check: $£180 + £240 = £420$ ✓

Worked Example: A recipe uses flour, butter and sugar in the ratio $5 : 3 : 2$. If the total mixture is 600 g, find the mass of each ingredient.

Total parts = $5 + 3 + 2 = 10$
1 part = $600 \div 10 = 60\,\text{g}$
Flour: $5 \times 60 = 300\,\text{g}$
Butter: $3 \times 60 = 180\,\text{g}$
Sugar: $2 \times 60 = 120\,\text{g}$

Worked Example: Alice and Bob share profits in the ratio $2 : 5$. Alice receives £140. How much does Bob receive?

Alice's 2 parts = £140, so 1 part = £70
Bob's share = $5 \times £70 = \mathbf{£350}$

Exam Tip: When only one share is given, find "1 part" first, then scale to the other share. You do not need to find the total first.

3.1.3 Comparing Using Ratios

Unit Ratio (1 : n or n : 1)

A unit ratio has 1 as one of its parts. It makes comparison easy.

Worked Example: Write $6 : 15$ as a unit ratio of the form $1 : n$.

Divide both by 6: $1 : 2.5$

Best Buy Problems

To find the best value, compare the **price per unit** (unit rate). Divide the price by the quantity for each option.

Worked Example: Box A: 400 g for £1.60. Box B: 550 g for £2.09. Which is the better value?

Box A: $£1.60 \div 400 = £0.0040$ per gram
Box B: $£2.09 \div 550 = £0.0038$ per gram
Box B is cheaper per gram — **Box B is better value**.

Exam Tip: Always state your conclusion clearly. Show the unit rate for each option so the examiner can award method marks even if arithmetic slips.

3.1.4 Direct Proportion

Two quantities are in **direct proportion** if, when one doubles, the other doubles (and so on). Their ratio remains constant. We write $y \propto x$, which means $y = kx$ for some constant k.

Finding the Constant of Proportionality

1. Use a known pair of values to find $k = y/x$.
2. Write the equation $y = kx$.
3. Use the equation to find unknown values.

Worked Example: y is directly proportional to x. When $x = 4$, $y = 20$. Find y when $x = 7$.

$k = y/x = 20/4 = 5$

Equation: $y = 5x$
When $x = 7$: $y = 5 \times 7 = \mathbf{35}$

Worked Example: 5 identical books cost £12.50. Find the cost of 8 books.

Cost per book = $£12.50 \div 5 = £2.50$
Cost of 8 books = $8 \times £2.50 = \mathbf{£20.00}$

Graph of Direct Proportion

A direct proportion graph is a **straight line through the origin**. The gradient equals k.

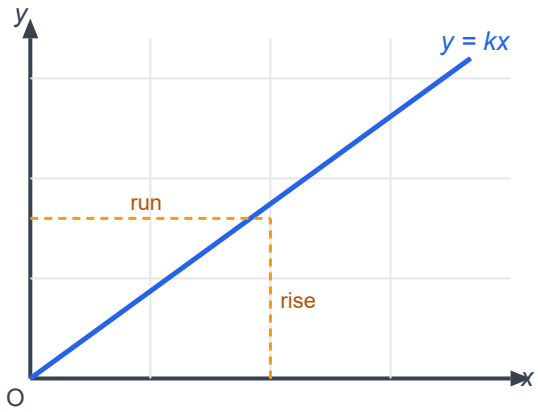

Fig 1 — Direct proportion: straight line through origin with gradient k

3.1.5 Inverse Proportion

Two quantities are in **inverse proportion** if, when one doubles, the other halves. We write $y \propto \frac{1}{x}$, which means $y = \frac{k}{x}$ for some constant k. The product xy is always constant.

Finding k for Inverse Proportion

1. Use a known pair of values: $k = xy$.
2. Write the equation $y = k/x$.
3. Substitute to find unknowns.

Worked Example: y is inversely proportional to x. When $x = 3$, $y = 8$. Find y when $x = 6$.

$k = xy = 3 \times 8 = 24$
Equation: $y = 24/x$
When $x = 6$: $y = 24/6 = \mathbf{4}$

Worked Example: 8 workers take 15 days to complete a project. How long would 12 workers take (assuming constant rate)?

$k = 8 \times 15 = 120$ (worker-days)
Time = $120 \div 12 = \mathbf{10}$ days

Higher: Other Proportionality Relationships Higher

Type	Notation	Equation
Direct (square)	$y \propto x^2$	$y = kx^2$
Direct (cube)	$y \propto x^3$	$y = kx^3$
Inverse (square)	$y \propto \frac{1}{x^2}$	$y = \frac{k}{x^2}$
Inverse (square root)	$y \propto \frac{1}{\sqrt{x}}$	$y = \frac{k}{\sqrt{x}}$

Worked Example: $y \propto x^2$. When $x = 3$, $y = 36$. Find y when $x = 5$.

$k = y/x^2 = 36/9 = 4$
Equation: $y = 4x^2$
When $x = 5$: $y = 4 \times 25 = \mathbf{100}$

Worked Example: $y \propto \frac{1}{x^2}$. When $x = 2$, $y = 5$. Find y when $x = 10$.

$k = yx^2 = 5 \times 4 = 20$
Equation: $y = 20/x^2$
When $x = 10$: $y = 20/100 = \mathbf{0.2}$

Proportion Graphs: Direct vs Inverse

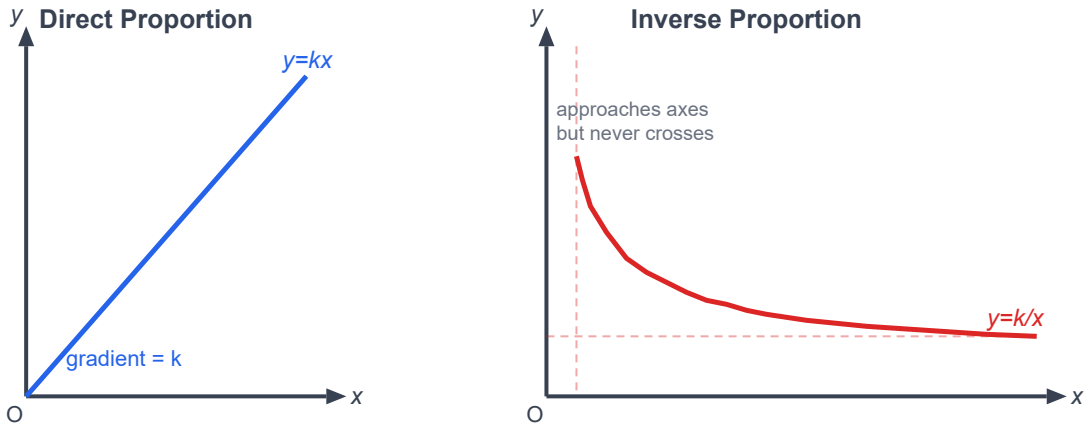

Fig 2 — Left: direct proportion ($y = kx$, straight line through origin). Right: inverse proportion ($y = k/x$, hyperbola approaching both axes).

Exam Tip: The inverse proportion graph never touches the axes. If asked to sketch it, draw a curve that gets closer and closer to each axis (asymptotic behaviour).

3.1.6 Proportionality in Geometry — Similar Shapes

Similar shapes have the same angles and all corresponding lengths are in the same ratio (the *scale factor* k). They are enlargements of each other.

Linear Scale Factor

If shape B is an enlargement of shape A with linear scale factor k, then every length in B is k times the corresponding length in A.

$$k = \frac{\text{length in image}}{\text{corresponding length in original}}$$

Area Scale Factor and Volume Scale Factor

If the linear scale factor is k, then:

$$\text{Area scale factor} = k^2$$

$$\text{Volume scale factor} = k^3$$

Relationship	Scale Factor
Lengths (1D)	k
Areas (2D)	k^2
Volumes (3D)	k^3

Worked Example: Two similar triangles have corresponding sides of 4 cm and 10 cm. (a) Find the area scale factor. (b) If the smaller triangle has area 24 cm^2, find the area of the larger.

Linear scale factor: $k = 10/4 = 2.5$
(a) Area scale factor: $k^2 = 2.5^2 = 6.25$
(b) Area of larger = $24 \times 6.25 = \mathbf{150}\,\mathrm{cm}^{\mathbf{2}}$

Worked Example: Two similar cylinders have heights 6 cm and 9 cm. The smaller has volume 80 cm^3. Find the volume of the larger.

Linear scale factor: $k = 9/6 = 1.5$
Volume scale factor: $k^3 = 1.5^3 = 3.375$
Volume of larger = $80 \times 3.375 = \mathbf{270}\,\mathrm{cm}^{\mathbf{3}}$

Worked Example: Two similar shapes have areas 50 cm² and 200 cm². Find the ratio of their perimeters.

Area scale factor: $200/50 = 4$
Linear scale factor: $k = \sqrt{4} = 2$
Ratio of perimeters = $1 : 2$

Exam Tip: Always identify what you are given — length, area or volume — and apply the correct power. If you are given areas and want lengths, take the square root. If you are given volumes and want lengths, take the cube root.

Formula Summary

Topic	Key Formula / Rule
Simplify ratio	Divide all parts by HCF; match units first
Divide in ratio $a : b$	1 part = Total $\div(a + b)$; shares = $a\times$part and $b\times$part
Direct proportion	$y = kx$; $k = y/x$ is constant
Inverse proportion	$y = k/x$; $k = xy$ is constant
$y \propto x^2$ (Higher)	$y = kx^2$; $k = y/x^2$
$y \propto 1/x^2$ (Higher)	$y = k/x^2$; $k = yx^2$
Similar shapes — area	Area SF $= k^2$
Similar shapes — volume	Volume SF $= k^3$

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